

ARUSHI SINGH

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EDUCATION

University of California, San Diego | La Jolla, CA

Bachelor of Science in Physics with Specialization in Astrophysics | June 2026

Minor in Computer Science | **Minor in Astronomy and Astrophysics**

Major GPA: 3.310/4.0 | Provost's Honors Roll (3x)

Society of Physics Students Outreach Board | **Association for Computing Machinery Member**

SELECTED RESEARCH EXPERIENCE

SPURS Scholar | May 2026 – September 2026

UC San Diego School of Physical Sciences | La Jolla, CA

Conducting full-time research on the gravitational n-body dynamics of globular cluster collisions

Performing academic paper reviews and creating and analyzing simulations using Cluster-Monte Carlo and REBOUND

Supervised by advisors Shahed Shayan-Arani and Dr. Kyle Kremer

Undergraduate Research Assistant | June 2025 – September 2025

USC Dornsife College of Letters, Arts and Sciences | Los Angeles, CA

UCI School of Physical Sciences | Irvine, CA

Conducting full-time research on Milky-Way mass galaxy accretion dynamics and galaxy formation and evolution using FIRE hydrodynamic simulations

Performing academic paper reviews and analyzing simulation data using Python

Supervised by advisors Patricia Fofie and Dr. James Bullock

Undergraduate Research Assistant | August 2024 – Present

UC San Diego Fuller Group | **UC San Diego Kremer Group** | La Jolla, CA

Researching primordial black hole (PBH) capture in our solar system using REBOUND

Investigating gravitational wave signatures from black hole-dominated globular clusters ("black clusters") using PeTar (C++ library)

Modeling black hole mergers and interactions in dense stellar environments with top-heavy initial mass functions

Exploring PBHs as dark matter candidates and potential explanations for astronomical anomalies

Supervised by Dr. George Michael Fuller and Dr. Kyle Kremer

Justice in Data Research Fellow | May 2024 – August 2024

The University of Texas at Arlington, SEUSI Lab | Remote

Completed two-week online bootcamp in large dataset management, FAIR data principles, and data processing techniques

Researched reliable nuclear energy and nuclear fuel recycling processes on a global scale

Created and presented scientific poster on application of data practices to research project

Supervised by Dr. Jessica Eisma

SELECTED PRESENTATIONS & CONFERENCE CONTRIBUTIONS

Conference Presentations

1. "Exploring Primordial Black Holes as a Dark Matter Candidate"

APS Global Physics Summit 2025 | Anaheim, CA | March 2025

Presented research poster on N-body simulations of primordial black hole capture in the solar system

Co-authors: Sean Mansour, Ishaan Satish, Shahed-Shayan Arani, George Fuller

2. "The Effects of AGN Feedback on Milky Way Mass Galaxy Formation"

Galfresca Meeting 2025 | La Jolla, CA | October 2025

Gave a talk on black hole and host galaxy coevolution and AGN feedback mechanisms in galaxies using FIRE-3 simulations

Co-authors: Patricia Fofie, James Bullock

3. **"The Effects of AGN Feedback on Milky Way Mass Galaxy Formation"**

APS Far West Section Meeting 2025 | Santa Cruz, CA | November 2025

Presented research poster on AGN feedback mechanisms in galaxy evolution using FIRE-3 simulations

Co-authors: Patricia Fofie, James Bullock

3. **"Using REBOUND to Model Globular Cluster Collisions "**

SPURS Poster Symposium | UC San Diego | August 2026

Presented research poster on the dynamics of globular cluster collisions using Cluster-Monte Carlo and REBOUND

Co-authors: Shasha Arani, Kyle Kremer

4. **"N-Body Simulations of Globular Cluster Collisions"**

Galfresca Meeting 2026 | Irvine, CA | September 2026

Gave a talk on the gravitational n-body dynamics of globular cluster collisions using Cluster-Monte Carlo and REBOUND

Co-authors: Shasha Arani, Kyle Kremer

Conference Planning

CU*IP 2025 (Conference for Undergraduate Gender Minorities in Physics) | UC San Diego | January 2025

Role: Undergraduate Planning Committee Member – Outreach, Advertising, and Budgeting Committees

SELECTED TEACHING & MENTORSHIP EXPERIENCE

Residence Assistant | August 2025 – June 2026

UC San Diego Sixth College | San Diego, CA

Providing referrals to campus resources and serving as a peer advisor to support 61 residents in Sixth College

Developing and implementing educational, cultural, and social programming to support student growth and success

Participating in duty rotation covering weekday evenings and weekend all-day shifts, responding to emergencies, mediating conflicts, and enforcing university policies to maintain a safe and inclusive living environment

Learning Assistant | March 2025 – June 2026

UC San Diego Department of Physics | San Diego, CA

Providing university-level teaching and assistant classroom instruction for lower-division physics courses

Conducting problem-solving sessions and office hours and providing individualized student support or tutoring when necessary

Instructional Assistant | March 2024 – September 2026

UC San Diego Department of Physics | San Diego, CA

Grading lab reports and providing guidance for lower-division physics laboratories

Meeting grading deadlines and working with a team to create a streamlined course for 300+ students at a time

Undergraduate Peer Advisor | August 2024 – September 2025

UC San Diego Jacobs School of Engineering | San Diego, CA

Providing academic advising to undergraduate and graduate students via email, phone, and in-person consultations

Assisting with admissions inquiries, CSE major/minor information, petitions, course enrollment, and campus resources

Creating informational materials (flyers, pamphlets, web content) for CSE programs

Performing administrative support including filing, textbook distribution, and organizational material development

Career Peer Education Program Intern | August 2023 – November 2024

UC San Diego Career Center | San Diego, CA

Peer-reviewed undergraduate resumes and cover letters to support internship and career exploration

Organized and executed career development events for the UCSD community

SELECTED HONORS & AWARDS

Provost's Honors Roll | UC San Diego

Fall 2022 - Fall 2025

Awarded for outstanding academic performance

Chair's Challenge Grant | UC San Diego Physics Department

Fall 2025

Competitive grant awarded to present research at APS Far West Section Meeting 2025

SPURS Scholarship Award | UC San Diego School of Physical Sciences

Summer 2026

Research grant awarded to students to conduct a project with a faculty mentor in Physical Science at UC San Diego

RELEVANT COURSEWORK

Physics & Astrophysics - Upper Division

PHYS 100A/B: Electromagnetism I/II

PHYS 110A/B: Mechanics I/II

PHYS 120: Circuits and Electronics

PHYS 130A/B: Quantum Mechanics I/II

PHYS 139: Special Topics - Machine Learning in Physics

PHYS 140A: Statistical and Thermal Physics I

PHYS 160: Stars and Compact Objects

PHYS 161: Black Holes

PHYS 199: Research for Undergraduates (4x)

PHYS 227: Cosmology

ASTR 150: Observational Optical Lab

ASTR 122: Physical Cosmology

ASTR 123: Galaxies

ASTR 211: Stellar Structure & Evolution

Computer Science

CSE 11: Accelerated Introduction to Programming

CSE 12: Basic Data Structures & Object-Oriented Design

CSE 20: Discrete Mathematics

CSE 21: Mathematics for Algorithms & Systems Analysis

CSE 29: Systems Programming and Software Tools

CSE 100: Advanced Data Structures

COGS 108: Data Science in Practice

CENG 15R: Engineering Computation Using MATLAB

TECHNICAL SKILLS

Programming Languages: Python, C++, C, MATLAB, Java

Scientific Computing: REBOUND (N-body simulations), PeTar (N-body simulations), FIRE/Gizmo (hydrodynamic simulations), CMC-COSMIC (orbit-averaging code for collisional stellar dynamics)

Software & Tools: Git, Jupyter Notebooks, Google Colab, LaTeX, Unix/Linux command line, NRP/Nautilus, Globus

Languages: English (Native), Hindi (Native), French (Conversational), Greek (Literate)

Interests: Computational astrophysics (dark matter probes, galaxy formation and evolution, black hole physics), gravitational waves, data science applications in physics using machine learning